

MITESH H. SHETTY

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Education:

Master of Science in Mechanical Engineering, University Of North Carolina at Charlotte	GPA 3.7/4.0	May/2019
Bachelor of Engineering in Mechanical Engineering, University of Mumbai, India	GPA 3.6/4.0	May/2016

Technical Skills:

- Programming languages: Python, Matlab.
- Machine learning: Octave, Spyder IDE, TensorFlow, Keras, NumPy, Pandas, Scikit-learn.
- Mechanical Softwares: CATIA, Solidworks, NX CAD, Ansys, CarSim, Adams, Lotus Shark, Simulink, Arduino, and LabVIEW.

Work Experience:

January/2018-December/2018

- Graduate Assistant:
 - Lectured & conducted experiments on PID motor position control, Temperature measurement sensors, microelectronic mechanical systems (MEMS), high pass filter, low pass filters to a batch of senior students.
 - Created a LabVIEW program that collects real-time data using National Instruments myDAQ assist.
 - Conducted lecture sessions on topics like Relative motion and acceleration, Particle kinetics, Work-energy methods, Impulse-momentum methods, Plane kinematics, Fixed axis rotation and general motion in the plane, Angular acceleration, Gear and pulley systems.
 - Lectured & conducted experiments on Electrostatics, Electric fields & potential, Capacitors, Current & magnetism, AC through LCR circuit & Optics to a batch of sophomore students.

Projects:

Data Science Projects | Python | Octave | Spyder IDE

January/2019-April/2019

- Developed an Artificial Neural Network for predicting whether a customer is fraudulent or not.
- Developed a Convolution Neural Network to predict whether the image provided is of a cat or a dog.
- Developed an unsupervised deep learning model to predict a customer's fraudulence using a self-organizing map.

Husqvarna Tilt & Telescope | Solidworks | Ansys

August/2018-May/2019

- Designed 3D CAD models & 2D drawings using GD&T for the tilt & telescoping subsystem.
- Performed manual Torque & torsional stress calculations on the entire steering assembly and individual components.
- Conducted FEA analysis on the tilt bracket & telescoping adapter using ANSYS to study & evaluate developed stress & deformation when subjected to max steering torque of 75ft-lbs.
- Conducted torque testing on the entire steering assembly by applying 75ft-lbs on the steering center bolt using a torque wrench.

Graduate projects | Matlab | Simulink

August/2017-May/2019

- Tire testing & modeling:
 1. Tested tires on a Michelin's M15 tire testing machine, gathered tire data was stored in a 4-D matrix.
 2. Generated a mathematical model of a tire utilizing the Pacejka equation.
 3. Studied the effects of parameters like Camber, Pressure, Vertical load and Lateral load on the performance of the tire.
- Determined the location of the center of gravity of a vehicle using the data obtained from a pendulum test.
- Simulated the longitudinal acceleration performance of a traction limited vehicle where the vehicle is to accelerate from rest, shifting gears as needed (no Torque from engine during shift), until a vehicle speed of 100 mph then the engine will be clutched and the driver will apply the specified force to the brake pedal slowing the car back to rest.
- Developed a path following algorithm for autonomous vehicles by implementing pure pursuit controller to study the interaction between a human driver & an autonomous vehicle. Vehicle dynamics & kinematics of a bicycle model for a four-wheeled vehicle were considered.
- Implemented a PID control system, on a Simulink model of a sail boat that adjusts the sail trim angle and the rudder angle.

All Terrain Vehicle | Catia | Solidworks | Ansys | MS Office

August/2014-April/2017

- Designed a 3-speed sequential gearbox with an open differential utilizing constant velocity joints for the ATV using CATIA and performed stress analysis using ANSYS.
- Constant Velocity (CV) Rzeppa joints were incorporated to transaxle end as well as the wheel end permitting a staggering ground clearance of 13".
- Custom splined attachments were designed & fabricated along the shafts which helped compensate changes in shaft length during wheel travel.
- Special case hardened steel adapters were designed & manufactured to join Malpezzi & CV joint.
- Built a custom rope brake dynamometer to test a 305cc 10hp Briggs and Stratton gasoline engine to generate power and torque curves.
- Tuned a CVTech continuously variable transmission with the Briggs and Stratton utilizing the data gathered from the dynamometer.
- Matched the CVT's engagement rpm with engine's maximum torque rpm and the shifting rpm with the engine's maximum power rpm by altering the tension & CAM weights inside the driving pulley as well as by altering the tension of the torsion spring fitted inside the driven spring.